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NCE Exam Practice Platform — Mauritius

2026

Information & Communication Technology

Grade 9 — National Certificate of Education (NCE)

PREDICTED PAPER 3 — Programming, Algorithms, Flowcharts & Internet Protocols

Specimen Practice Paper — Not an Official MES Document

Duration: 1 hour 45 minutes | **Total Marks:** 100

Candidate Name: _____

Class / School: _____

INSTRUCTIONS TO CANDIDATES

1. Write your name and class in the spaces provided above.
2. Answer **ALL** questions in all three sections.
3. Section A: Tick **ONE** box only for each multiple-choice question.
4. Section B: Write your answers in the spaces provided.
5. Section C: Use additional paper if the space provided is insufficient.
6. Calculators are **NOT** permitted.
7. Flowcharts may be drawn using pencil; all other answers in ink.

Section	Topic	Marks
A	Multiple Choice Questions	25
B	Short Structured Questions	40
C	Extended / Scenario Questions	35
TOTAL		100

SECTION A — Multiple Choice Questions [25 marks]

Tick the ONE correct answer for each question. Each question is worth **1 mark**.

1. An *algorithm* is best defined as:

1

- ☐ A a high-level programming language
- ☐ B a set of step-by-step instructions to solve a problem
- ☐ C a diagram showing data flow through a system
- ☐ D a list of variables used in a program

2. In a flowchart, a *diamond* shape represents:

1

- ☐ A the start or end of the process
- ☐ B a process or calculation step
- ☐ C a decision (yes/no question)
- ☐ D an input or output operation

3. What is the output of the following pseudocode when $x = 5$?

1

```
IF  $x > 3$  THEN
  OUTPUT  $x * 2$ 
ELSE
  OUTPUT  $x + 1$ 
ENDIF
```

- ☐ A 6
- ☐ B 8
- ☐ C 10
- ☐ D 5

4. A *FOR loop* is most suitable when:

1

- ☐ A the number of iterations is unknown in advance
- ☐ B the loop must run at least once before checking a condition
- ☐ C the number of iterations is known in advance
- ☐ D a condition is checked before each iteration with no defined limit

5. Which of the following is the correct Python syntax to display the text "Hello, World!" on screen?

1

- ☐ A `PRINT("Hello, World!")`
- ☐ B `print("Hello, World!")`
- ☐ C `output "Hello, World!"`

☐ D `display("Hello, World!")`

6. What is the value of `total` after this Python code runs?

1

```
total = 0
for i in range(1, 5):
    total = total + i
```

- ☐ A 10
- ☐ B 15
- ☐ C 4
- ☐ D 14

7. In programming, a *variable* is:

1

- ☐ A a fixed value that cannot change during program execution
- ☐ B a named memory location that stores a value which can change
- ☐ C a predefined function in Python
- ☐ D the result of a Boolean expression

8. Which data type would be most appropriate to store a student's *age*?

1

- ☐ A String
- ☐ B Boolean
- ☐ C Integer
- ☐ D Float

9. A *function* (or subprogram) is used in programming to:

1

- ☐ A connect to a database
- ☐ B group reusable code that can be called multiple times
- ☐ C declare global variables
- ☐ D terminate a loop

10. In pseudocode, the structure that tests a condition and repeats while it is TRUE is:

1

- ☐ A FOR ... NEXT
- ☐ B IF ... ENDIF
- ☐ C WHILE ... ENDWHILE
- ☐ D CASE ... ENDCASE

11. Which flowchart symbol is used for *start* and *end*?

1

- ☐ A Rectangle
- ☐ B Diamond

- ☐ C Parallelogram
- ☐ D Rounded rectangle (oval/capsule)

12. In Python, the command `input()` always returns data as which type? 1

- ☐ A Integer
- ☐ B Float
- ☐ C String
- ☐ D Boolean

13. What is the purpose of *test data* when checking an algorithm? 1

- ☐ A To make the program run faster
- ☐ B To verify that the algorithm produces correct results for given inputs
- ☐ C To compress the code
- ☐ D To document variable names

14. A *syntax error* in a program occurs when: 1

- ☐ A the program produces incorrect output
- ☐ B the program runs but crashes at a particular line
- ☐ C the program's code violates the grammar rules of the programming language
- ☐ D the program gives a logically wrong answer

15. The *Internet* is best described as: 1

- ☐ A a single large network owned by one organisation
- ☐ B a global network of interconnected networks using TCP/IP protocols
- ☐ C the same thing as the World Wide Web
- ☐ D a network limited to one country

16. FTP (File Transfer Protocol) is used to: 1

- ☐ A send and receive emails
- ☐ B display web pages in a browser
- ☐ C transfer files between computers over a network
- ☐ D assign IP addresses to devices automatically

17. SMTP (Simple Mail Transfer Protocol) is used for: 1

- ☐ A receiving emails from a mail server
- ☐ B sending emails from a client to a mail server
- ☐ C encrypting web traffic
- ☐ D resolving domain names to IP addresses

18. Packet switching works by: 1
- ☐ A sending all data through one dedicated path between sender and receiver
 - ☐ B breaking data into small packets that travel independently and are reassembled at the destination
 - ☐ C compressing data before transmission
 - ☐ D encrypting all data before it leaves the sender
19. What does a *URL* (Uniform Resource Locator) identify? 1
- ☐ A A user's login credentials on a website
 - ☐ B The specific address of a resource on the internet
 - ☐ C A packet's routing path through the internet
 - ☐ D The speed of a network connection
20. Which protocol automatically assigns an IP address to a device when it joins a network? 1
- ☐ A DNS
 - ☐ B FTP
 - ☐ C DHCP
 - ☐ D SMTP
21. In a linear search of a list with 10 items, what is the *maximum* number of comparisons needed? 1
- ☐ A 1
 - ☐ B 5
 - ☐ C 9
 - ☐ D 10
22. A *binary search* requires the list to be: 1
- ☐ A sorted in ascending or descending order
 - ☐ B stored in a 2D array
 - ☐ C in binary (0s and 1s)
 - ☐ D no longer than 100 items
23. Which of the following is an example of a *logic error*? 1
- ☐ A Forgetting a closing bracket in Python
 - ☐ B Using `>` instead of `>=` in a condition, causing an off-by-one error
 - ☐ C Dividing by zero at runtime
 - ☐ D Misspelling a keyword such as `prit` instead of `print`

24. What is the World Wide Web (WWW)?

1

- ☐ A A physical network of cables that connects computers globally
- ☐ B A collection of web pages and resources accessed via the internet using HTTP/HTTPS
- ☐ C A protocol for sending emails
- ☐ D Another name for the internet

25. A *bubble sort* algorithm works by:

1

- ☐ A finding the smallest element and placing it at the front in each pass
- ☐ B dividing the list in half and sorting each half recursively
- ☐ C repeatedly comparing adjacent pairs and swapping them if they are in the wrong order
- ☐ D searching for a target by checking the middle element first

SECTION B — Short Structured Questions [40 marks]

Answer **ALL** questions.

B1. Trace Table

5

Study the following pseudocode:

```
count ← 0
total ← 0
FOR i ← 1 TO 5
    total ← total + i
    IF i MOD 2 = 0 THEN
        count ← count + 1
    ENDIF
NEXT i
OUTPUT total, count
```

(a) Complete the trace table for the pseudocode above.[3]

i	total	count	i MOD 2 = 0?
1			
2			
3			
4			
5			

(b) State the final values output by the pseudocode.[1]

(c) State what the variable count is counting.[1]

B2. Writing Pseudocode

5

(a) Write pseudocode for an algorithm that: (i) asks the user to input their exam mark (0–100); (ii) outputs "Distinction" if mark ≥ 80 , "Pass" if mark ≥ 50 , and "Fail" otherwise. [4]

-
- (b) Give one example of *normal* test data and one example of *boundary* test data for this algorithm. [1]
-

B3. Python Programming

5

Study the Python code below:

```
scores = [45, 78, 62, 91, 55]
total = 0
for s in scores:
    total = total + s
average = total / len(scores)
print("Average:", average)
```

- (a) State what the variable *scores* is.[1]
-

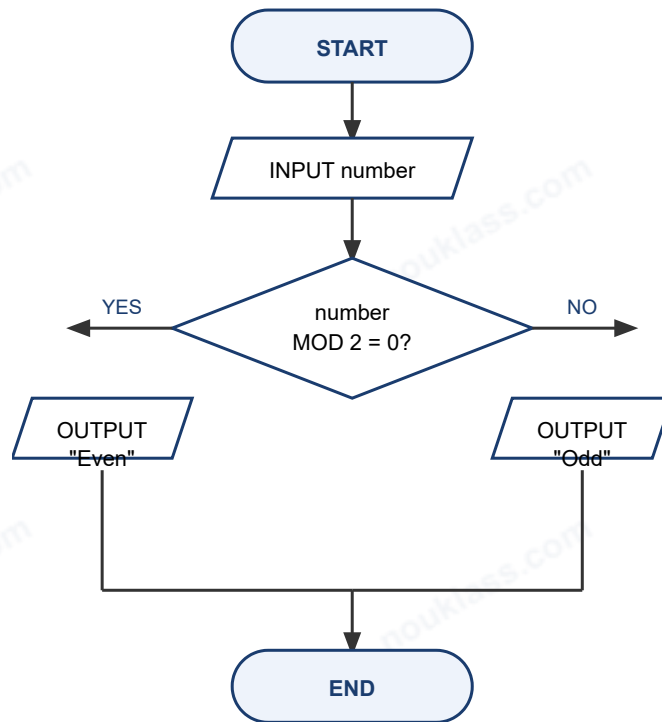
- (b) Calculate and state the output of this program. Show your working.[2]

- (c) Modify the code to also output the *highest* score in the list. Write the additional line(s) needed. [2]

B4. Flowcharts

5

The flowchart below describes an algorithm. Study it carefully and answer the questions.



(a) State what this algorithm does.[1]

(b) What is the output when the user inputs the number 7?[1]

(c) Name the flowchart symbol used for **INPUT number** and the symbol used for **MOD 2 = 0?** [2]

(d) Suggest a modification to make this algorithm check *multiple numbers* until the user inputs 0. State which control structure you would add. [1]

B5. Internet Protocols

5

(a) Match each protocol to its correct function by writing the letter in the table. [4]

Protocol	Function (write letter)
HTTP	
FTP	

SMTP	
DHCP	

- A** — Automatically assigns IP addresses to devices on a network
B — Transfers files between computers
C — Sends web page requests and receives responses
D — Sends email messages from client to server

(b) Explain one difference between the Internet and the World Wide Web. [1]

B6. Sorting and Searching

5

(a) The list below is to be sorted using **bubble sort**. Show the state of the list after the *first full pass*. [2]

Original list: 8 3 1 6 4

(b) A binary search is performed on the sorted list: 2 5 8 11 15 20 24 30. The target is **20**. Show the steps of the search, including the midpoint checked at each step. [2]

(c) Give *one* advantage of binary search over linear search. [1]

B7. Python — Functions

5

(a) State *two* benefits of using functions (subprograms) in a large program. [2]

(b) Write a Python function called `is_pass` that takes a mark as a parameter and returns `True` if the mark is 50 or above, and `False` otherwise. [2]

```
def is_pass(mark):
```

-
- (c) Write a line of Python code that calls the function `is_pass` with the argument 65 and prints the returned value. [1]
-

B8. Debugging

5

The following Python code contains **three errors**. Identify and correct each error.

```
name = input("Enter your name: ")
age = input("Enter your age: ")
If age >= 18
    print("Welcome, " + name)
else
    print("Sorry, you must be 18 or older")
```

- (a) List the **three errors** and state what type of error each is (syntax, logic, or runtime). [3]
-
-
-

- (b) Write the corrected version of the code.[2]

SECTION C — Extended / Scenario Questions [35 marks]

C1. Scenario: Student Grade Calculator

14

A teacher wants a program that reads in a student's marks for 5 subjects, calculates the average, and outputs a final grade: **Distinction** (average ≥ 80), **Merit** (average ≥ 65), **Pass** (average ≥ 50), or **Fail** (average < 50). The program must also display the highest mark achieved.

- (a) Write pseudocode for the complete algorithm described above. Your pseudocode must include: input, a loop, selection (IF/ELSEIF/ELSE), calculation of the average and highest mark, and output. [6]

- (b) In the space below, draw a flowchart for the *grade classification* part only (from "average calculated" to "output grade"). Use proper flowchart symbols. Your flowchart must show all four possible outcomes. [5]

Draw your flowchart here — use diamond shapes for decisions, rectangles for processes, parallelograms for output



- (c) Complete the trace table for the pseudocode below, using the inputs: 72, 85, 60, 78, 90. [3]

```
total ← 0
highest ← 0
FOR i ← 1 TO 5
    INPUT mark
    total ← total + mark
    IF mark > highest THEN
        highest ← mark
    ENDIF
NEXT i
average ← total / 5
```

i	mark	total	highest
1	72		
2	85		
3	60		
4	78		
5	90		

Final average = _____ Final grade output = _____

C2. Scenario: How a Web Page Loads — Internet Protocols in Action

11

Meera opens her laptop and types **https://library.nouklass.com/books** into her browser. Her laptop is connected to the school's LAN which has a router connecting to the internet via a fibre line.

- (a) Identify the **protocol**, **domain name**, and **path** in the URL *https://library.nouklass.com/books*.

[3]

- (b) Describe the role of **DNS** in loading this web page, including what query is made and what is returned.

[3]

- (c) Explain how **packet switching** is used when the web page data travels from the server to Meera's browser. Include: how data is broken up, how packets travel, and how they are reassembled.

[3]

- (d) Explain why HTTPS is used rather than HTTP for this library website. What does the 'S' add?

[2]

C3. Python Programming — Extended

10

- (a) Write a complete Python program that: asks the user to enter 6 integers; stores them in a list; outputs the sum, the maximum, and the minimum of the list. [5]

- (b) Study the Python function below:[3]

```
def count_vowels(word):  
    vowels = "aeiouAEIOU"  
    count = 0  
    for letter in word:  
        if letter in vowels:  
            count = count + 1  
    return count  
  
result = count_vowels("Mauritius")  
print(result)
```

What is the output of this program? Show your working by listing which letters are vowels in "Mauritius".

- (c) The function above uses iteration (a for loop). Explain why iteration is appropriate here rather than writing out separate if statements for each letter in the word. [2]

MARK SCHEME — Paper 3

Accept any pseudocode that correctly expresses the algorithm. Python answers accepted if logic is correct and syntax would work in Python 3. Flowchart diagrams: award marks for correct structure and symbols, allow minor labelling variation.

SECTION A — Multiple Choice

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	C	3	C	4	C	5	B
6	A	7	B	8	C	9	B	10	C
11	D	12	C	13	B	14	C	15	B
16	C	17	B	18	B	19	B	20	C
21	D	22	A	23	B	24	B	25	C

SECTION B

Q	Expected Answer	Marks
B1(a)	i=1: total=1, count=0, No i=2: total=3, count=1, Yes i=3: total=6, count=1, No i=4: total=10, count=2, Yes i=5: total=15, count=2, No [1 per two correct rows]	3
B1(b)	total = 15, count = 2	1
B1(c)	The number of even integers in the range 1 to 5	1
B2(a)	INPUT mark [1]; IF mark $\geq$ 80 THEN OUTPUT "Distinction" [1]; ELSEIF mark $\geq$ 50 THEN OUTPUT "Pass" [1]; ELSE OUTPUT "Fail" ENDIF [1]. Award 4 for complete correct logic in any valid pseudocode style.	4
B2(b)	Normal: e.g. 75 (expected Pass output) [0.5]; Boundary: 80 or 50 (at boundary between grades) [0.5] — award 1 if both given correctly	1
B3(a)	A list (array) of integers	1
B3(b)	$45+78+62+91+55 = 331$; $331/5 = 66.2$; Output: Average: 66.2 [1 for working, 1 for 66.2]	2
B3(c)	print("Highest:", max(scores)) OR equivalent using a loop to find the maximum [1 for correct approach, 1 for correct syntax]	2
B4(a)	Determines whether a number is even or odd	1
B4(b)	Odd	1
B4(c)	INPUT number: parallelogram [1]; MOD 2 = 0?: diamond [1]	2
B4(d)	Add a WHILE loop (or DO-WHILE) that repeats the process while the number entered is not 0	1
B5(a)	HTTP: C; FTP: B; SMTP: D; DHCP: A [1 per correct pair — 4 marks]	4

B5(b)	The Internet is the physical global network of interconnected networks [1]; the WWW is a collection of web pages and resources accessed via the internet using HTTP [1] — accept any answer that correctly distinguishes infrastructure from service	1
B6(a)	After first pass: 3 1 6 4 8 [1 for correct sequence; 1 for evidence of swap steps]	2
B6(b)	Mid = index 4 = 15; 20 > 15 so search right half [1]; Mid = index 6 = 24; 20 < 24 so search left; Mid = index 5 = 20; found [1]	2
B6(c)	Binary search is faster on large sorted lists / fewer comparisons needed on average ($O(\log n)$ vs $O(n)$)	1
B7(a)	Any 2 of: code reuse / avoids repetition; easier to test and debug; improves readability; easier to maintain [1 each]	2
B7(b)	def is_pass(mark): / if mark >= 50: return True / else: return False (or return mark >= 50) [1 correct def/param, 1 correct logic]	2
B7(c)	print(is_pass(65))	1
B8(a)	Error 1: If should be if (lowercase) — syntax [1]; Error 2: Missing colon after age >= 18 — syntax [1]; Error 3: age is a string (from input), must convert with int() before comparing — runtime/type error [1]	3
B8(b)	age = int(input(...)); if int(age) >= 18: (with colon) / else: (with colon) — both corrected [1 per correction, max 2]	2

SECTION C

Q	Expected Answer / Mark Points	Marks
C1(a)	INPUT loop for 5 marks [1]; accumulate total and track highest [1]; calculate average = total / 5 [1]; IF/ELSEIF for Distinction/Merit/Pass/Fail with correct boundaries [2]; OUTPUT grade and highest [1]	6
C1(b)	START/END ovals [1]; decision diamonds for each threshold (≥ 80 , ≥ 65 , ≥ 50) with correct YES/NO paths [2]; output shapes for each grade [1]; correctly connected arrows [1]	5
C1(c)	i=1: total=72, highest=72 i=2: total=157, highest=85 i=3: total=217, highest=85 i=4: total=295, highest=85 i=5: total=385, highest=90 [2 for correct table]; average=385/5=77.0 [0.5]; grade = Merit [0.5]	3
C2(a)	Protocol: https [1]; Domain: library.noukclass.com [1]; Path: /books [1]	3
C2(b)	Browser sends a DNS query with the domain name library.noukclass.com [1]; DNS server looks up the domain in its records [1]; returns the corresponding IP address to the browser [1]	3
C2(c)	Web page data is broken into small packets [1]; each packet travels independently through different routers and may take different routes [1]; TCP reassembles packets at the destination in the correct order and requests retransmission of any lost packets [1]	3

C2(d)	HTTPS uses SSL/TLS encryption [1]; all data between Meera's browser and the server is encrypted, protecting login details and personal data from interception [1]	2
C3(a)	numbers = [] and a loop using append/input [1]; input inside loop for 6 values [1]; sum() or accumulator [1]; max() and min() [1]; print all three [1]	5
C3(b)	Vowels in "Mauritius": M-no, a-yes, u-yes, r-no, i-yes, t-no, i-yes, u-yes, s-no = 5 vowels; Output: 5 [2 for correct identification + output, 1 for output only]	3
C3(c)	Iteration handles words of any length without needing to know the length in advance [1]; writing separate if statements would only work for a fixed-length word and would not generalise [1]	2